7th, KM Haridwar, National Highway Vardhmanpuram,
Roorkee, Rehmadpur, Uttarakhand 247667

Lab Sheet-06

Python Programming Lab – Set Operations

MCA I Semester (Session 2026-2027)

Objectives:

- To understand the use of sets in Python for handling unique, unordered collections of data.
- To implement fundamental and advanced set operations such as union, intersection, difference, and membership testing.

Requirements:

- Python 3.x installed on the system (IDLE, PyCharm, VS Code, or any preferred IDE).
- Basic knowledge of Python data types and loops.
- Understanding of mathematical set operations and user input/output handling.

Program List:

1. Write a Python program to create and display a set.
2. Write a program to create a set with different data types.
3. Write a program to add a single element to a set.
4. Write a program to add multiple elements to a set.
5. Write a program to remove an element from a set using remove().
6. Write a program to remove an element from a set using discard().
7. Write a program to clear all elements from a set.
8. Write a program to check if an element exists in a set.
9. Write a program to find the length of a set.
10. Write a program to iterate through a set and print its elements.
11. Write a program to find the union of two sets.
12. Write a program to find the intersection of two sets.
13. Write a program to find the difference of two sets.
14. Write a program to find the symmetric difference of two sets.
15. Write a program to check if a set is a subset of another set.
16. Write a program to check if a set is a superset of another set.
17. Write a program to check if two sets are disjoint.

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18. Write a program to remove all elements of one set from another set.
19. Write a program to create the power set of a given set.
20. Write a program to find the Cartesian product of two sets.
21. Write a program to count distinct elements from a list using a set.
22. Write a program to find duplicate elements in a list using a set.
23. Write a program to remove duplicates from a list using a set.
24. Write a program to convert a string into a set of characters.
25. Write a program to find common letters in two strings using sets.
26. Write a program to find unique words in a sentence using a set.
27. Write a program to check if two strings are anagrams using sets.
28. Write a program to create a frozen set and demonstrate its properties.
29. Write a program to find maximum and minimum values in a set.
30. Write a program to perform all basic set operations (union, intersection, difference, symmetric difference) in one program.

Instruction:

- Save each program in a separate .py file (e.g., Q1_set_creation.py).
- Use **user input** wherever possible (e.g., reading elements from input).
- Add **comments** in code to explain the logic.
- Display results in a **clear, readable format**.
- Test the programs with different inputs before submission.

Guideline:

- Use **set operations only** (avoid unnecessary conversions to lists unless specified).
- Do not use advanced libraries unless required.
- Follow **Python indentation (4 spaces)** and coding standards.
- Validate user inputs when necessary.
- Ensure programs are free of syntax/runtime errors.

Submission:

- Submit the source code files (.py) for each task along with a brief report documenting the implementation details and the results of the executions.



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